WEEK 5 DEEP LEARNING PROJECT

Diabetes Prediction Using Deep Learning

Binary Classification using a PyTorch Multi-Layer Perceptron

Item	Details
Project Type	Deep Learning Classification
Dataset	CDC Diabetes Health Indicators
Dataset Size	253,680 original records
Features	21 input features
Target	Diabetes_binary
Framework	PyTorch
Model	Multi-Layer Perceptron (MLP)
Training Device	CPU

Prepared as a practical deep learning project

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1. Executive Summary

This project develops a binary diabetes prediction system using a feed-forward neural network implemented in PyTorch. The objective is to learn nonlinear relationships between demographic, lifestyle, health-status and socioeconomic indicators and the binary diabetes target.

The project uses the CDC Diabetes Health Indicators dataset, containing 253,680 records and 22 columns, including 21 predictor variables and the target variable Diabetes_binary.

The analysis first investigated data quality, class imbalance, duplicate observations, feature distributions, correlations, class-wise differences and potential IQR-based outliers. These analyses were performed before model development so that preprocessing and modeling decisions were evidence-based.

After removing exact duplicate rows, the dataset contained 229,474 observations. A stratified train/validation/test split was used, and StandardScaler was fitted only on the training data to avoid data leakage.

The final PyTorch MLP achieved a test accuracy of 70.48%, precision of 31.54%, recall of 79.49%, F1-score of 45.17% and ROC-AUC of 0.8167 at the 0.50 classification threshold. The high recall reflects the use of a weighted binary cross-entropy loss to address the imbalanced target distribution.

2. Problem Statement

Diabetes prediction is a binary classification problem in which the model estimates whether an individual belongs to a diabetes-positive or diabetes-negative class based on available health and demographic indicators.

A major challenge in this dataset is class imbalance. Only 13.93% of the original observations belong to the diabetes-positive class, while 86.07% belong to the non-diabetes class. Therefore, accuracy alone is not sufficient for evaluating model performance.

The project therefore emphasizes precision, recall, F1-score, ROC-AUC, confusion-matrix analysis and threshold sensitivity in addition to accuracy.

3. Objectives

- Understand and explore the CDC Diabetes Health Indicators dataset.
- Identify data-quality issues before model training.
- Quantify duplicate records and their impact on the target distribution.
- Investigate potential statistical outliers using the IQR method.
- Analyze differences between diabetes and non-diabetes groups.
- Identify features associated with the diabetes target.
- Build an end-to-end preprocessing pipeline.
- Develop a neural-network classifier using PyTorch.
- Address class imbalance using weighted BCEWithLogitsLoss.
- Evaluate the model using multiple classification metrics.
- Analyze classification errors using a confusion matrix.
- Investigate the effect of probability thresholds.
- Document limitations and possible future improvements.

4. Dataset Description

The project uses the CDC Diabetes Health Indicators dataset from the UCI Machine Learning Repository. The dataset contains health-related survey indicators and demographic or socioeconomic variables.

The original dataset contains 253,680 rows and 22 columns. There are 21 input features and one binary target variable, Diabetes_binary.

Property	Value

Original observations	253,680
Original columns	22
Predictor features	21
Target variable	Diabetes_binary
Missing values	0
Exact duplicate rows	24,206
Duplicate percentage	9.54%
Rows after duplicate removal	229,474
Original non-diabetes class	218,334 (86.07%)
Original diabetes class	35,346 (13.93%)

The target variable is binary: 0 represents no diabetes and 1 represents diabetes.

5. Data Quality and Exploration

5.1 Data Types and Missing Values

All 22 columns were stored as int64 values. No missing values were detected in the dataset. Because the variables are encoded survey indicators and ordinal categories, integer storage does not imply that all variables are continuous.

5.2 Feature Cardinality

The dataset contains a mixture of binary, ordinal and bounded numeric variables. Fifteen variables are binary-like, while variables such as BMI, MentHlth and PhysHlth have larger numbers of unique values.

Feature Group	**Examples**	**Interpretation**
Binary	HighBP, HighChol, Smoker, Sex	0/1 indicators
Bounded numeric	BMI	Continuous-like health measure
Health scale	GenHlth	Five-level ordinal scale
Health-day counts	MentHlth, PhysHlth	0–30 day counts
Ordinal	Age, Education, Income	Ordered categories

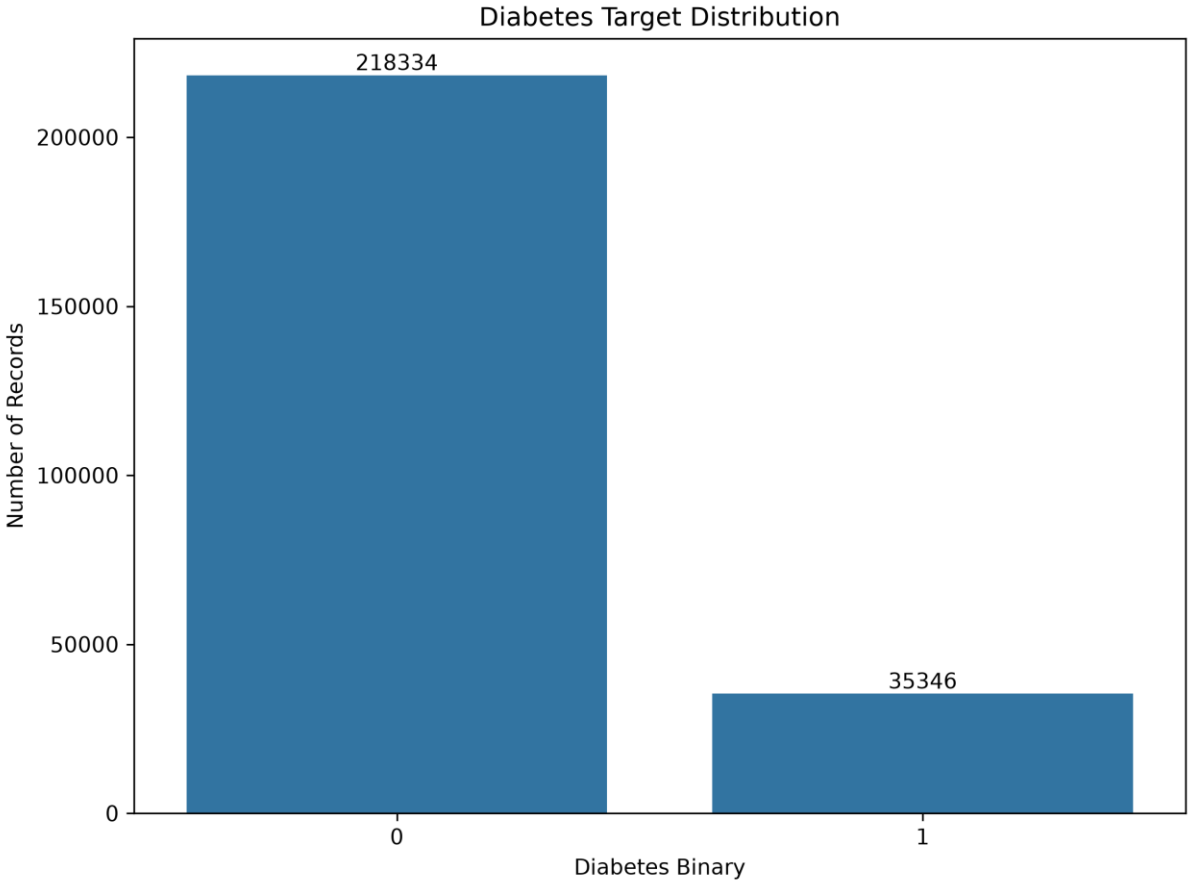


Figure 1. Original target distribution showing the strong class imbalance.

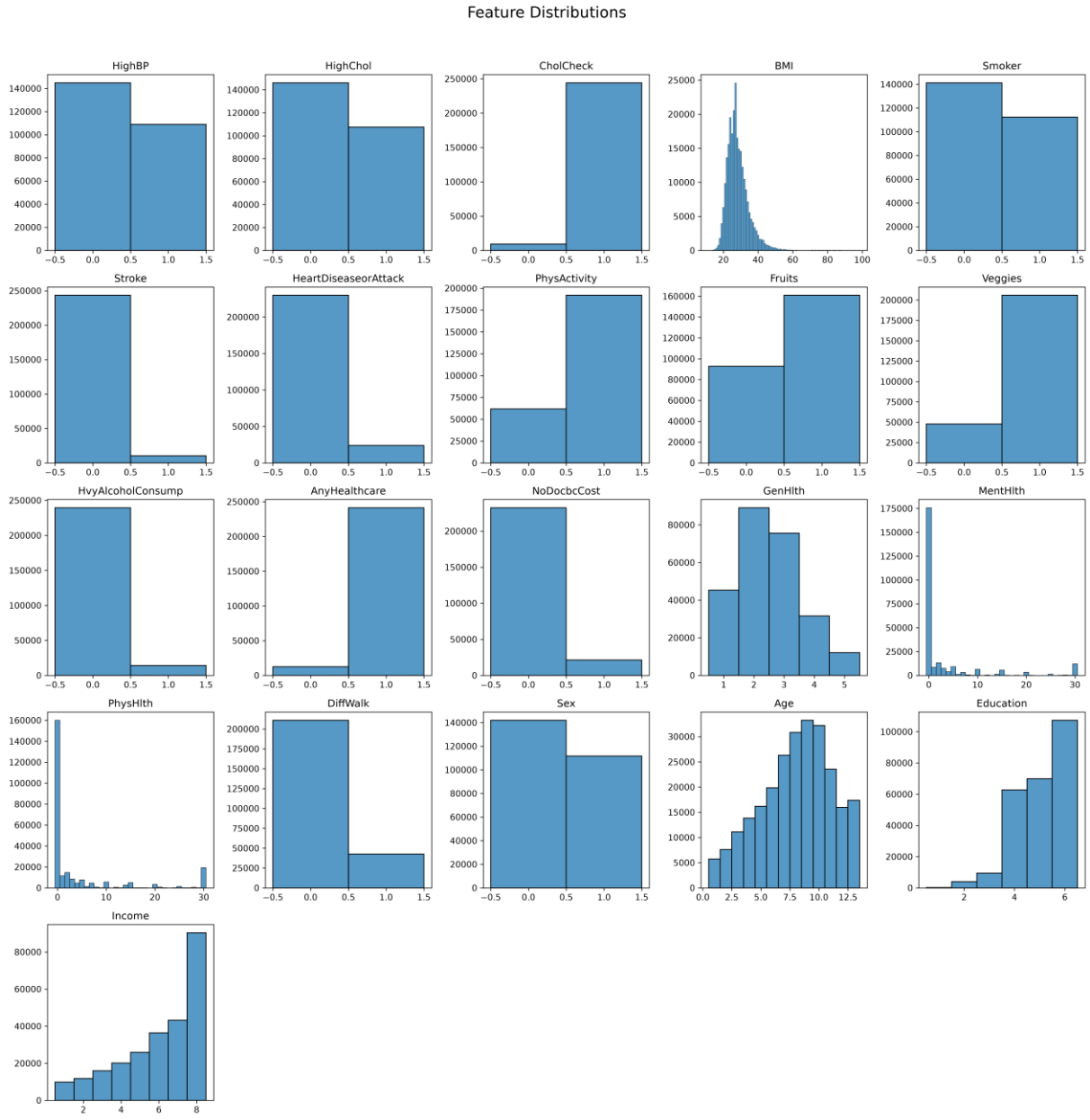


Figure 2. Feature distributions across the dataset.

6. Duplicate Analysis

Exact duplicate analysis identified 24,206 duplicated rows, equal to approximately 9.54% of the original dataset. Because exact duplicates do not provide additional independent information, they were removed before model splitting.

This step is important because duplicate records can cause the same observation or an identical observation to appear across different dataset partitions, potentially producing overly optimistic evaluation results.

Measure	**Value**
Original rows	253,680
Duplicate rows	24,206
Duplicate percentage	9.54%
Rows retained	229,474
Rows removed	24,206

The class distribution also changed slightly after duplicate removal. The proportion of the no-diabetes class decreased by approximately 1.36 percentage points, while the diabetes class proportion increased by approximately 1.36 percentage points.

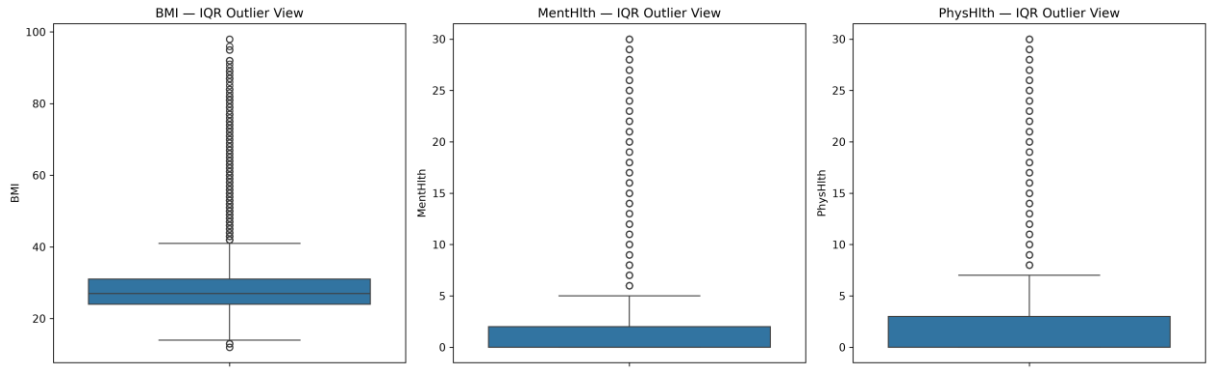


Figure 3. IQR-based statistical outlier analysis for selected numerical features.

7. Outlier Analysis and Impact Assessment

Potential outliers were investigated using the Interquartile Range (IQR) method. An observation was flagged when it was below Q1 − 1.5×IQR or above Q3 + 1.5×IQR.

The purpose of this analysis was not to automatically delete every statistical outlier. Instead, the flagged values were examined in the context of the meaning and valid range of each variable.

Feature	Q1	Q3	IQR	Lower Bound	Upper Bound	Potential Outliers
BMI	24	31	7	13.5	41.5	9,847 (3.882%)
MentHlth	0	2	2	-3	5	36,208 (14.273%)
PhysHlth	0	3	3	-4.5	7.5	40,949 (16.142%)

7.1 BMI Impact Assessment

BMI has a Q1 of 24 and Q3 of 31, resulting in an IQR of 7. The IQR rule flags values below 13.5 or above 41.5. A total of 9,847 observations were flagged. However, BMI values ranged from 12 to 98, and extreme values may represent genuine observations rather than data-entry errors.

7.2 Mental and Physical Health Days

MentHlth and PhysHlth are bounded survey variables representing the number of days in a recent period. Their distributions are heavily concentrated near zero, which makes the IQR rule flag many valid higher values. Therefore, removing these observations solely because they are statistically unusual would risk deleting meaningful health information.

7.3 Final Outlier Decision

No observations were removed solely because they were identified as IQR outliers. The duplicate-removal step was retained because exact duplicates represent repeated records rather than unusual but potentially valid observations.

8. Exploratory Data Analysis

Exploratory analysis was expanded beyond basic distributions to include target relationships, class-wise comparisons, prevalence analysis and correlations. This provides concrete evidence for the subsequent modeling decisions.

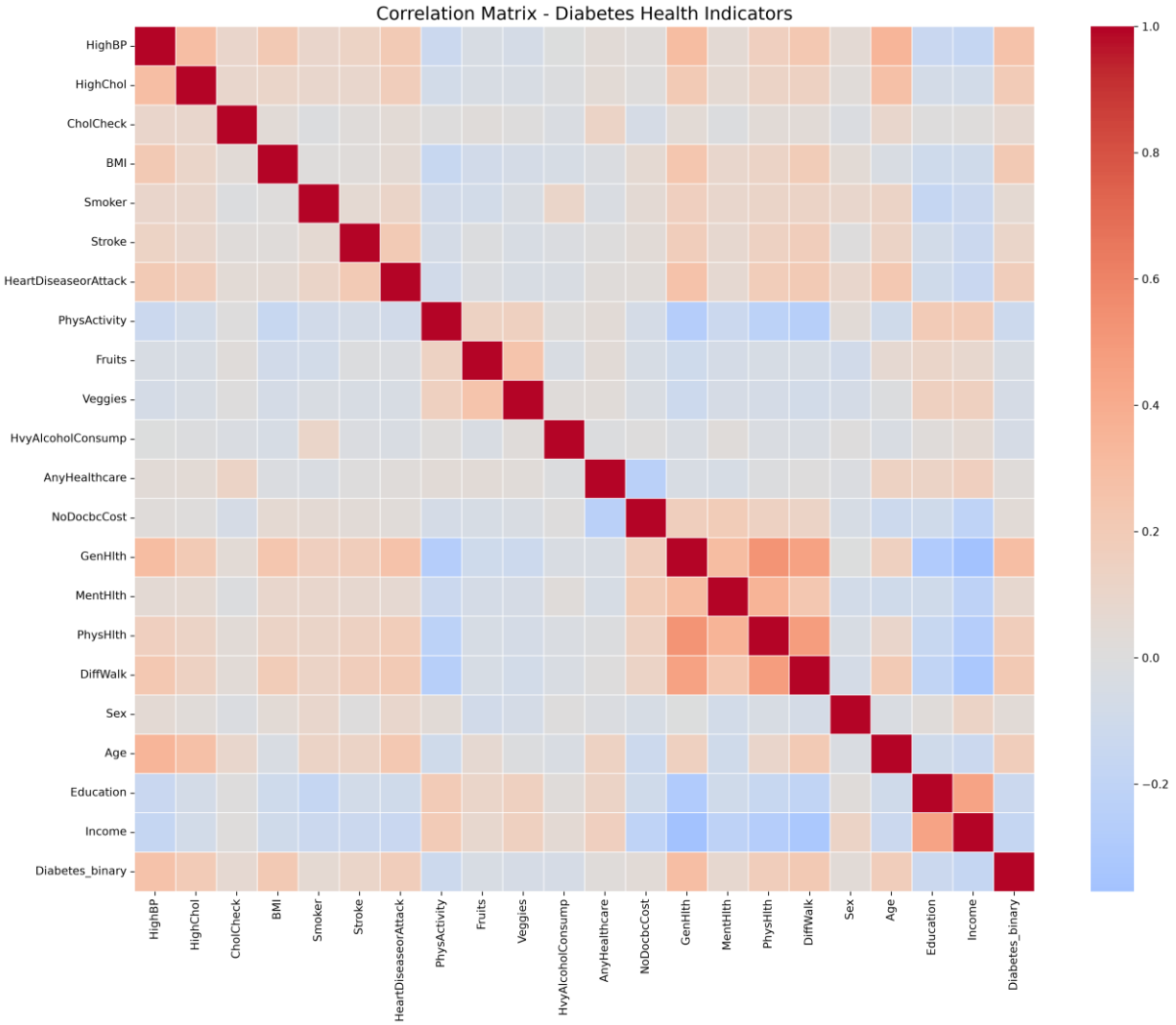


Figure 4. Correlation heatmap across the encoded dataset features.

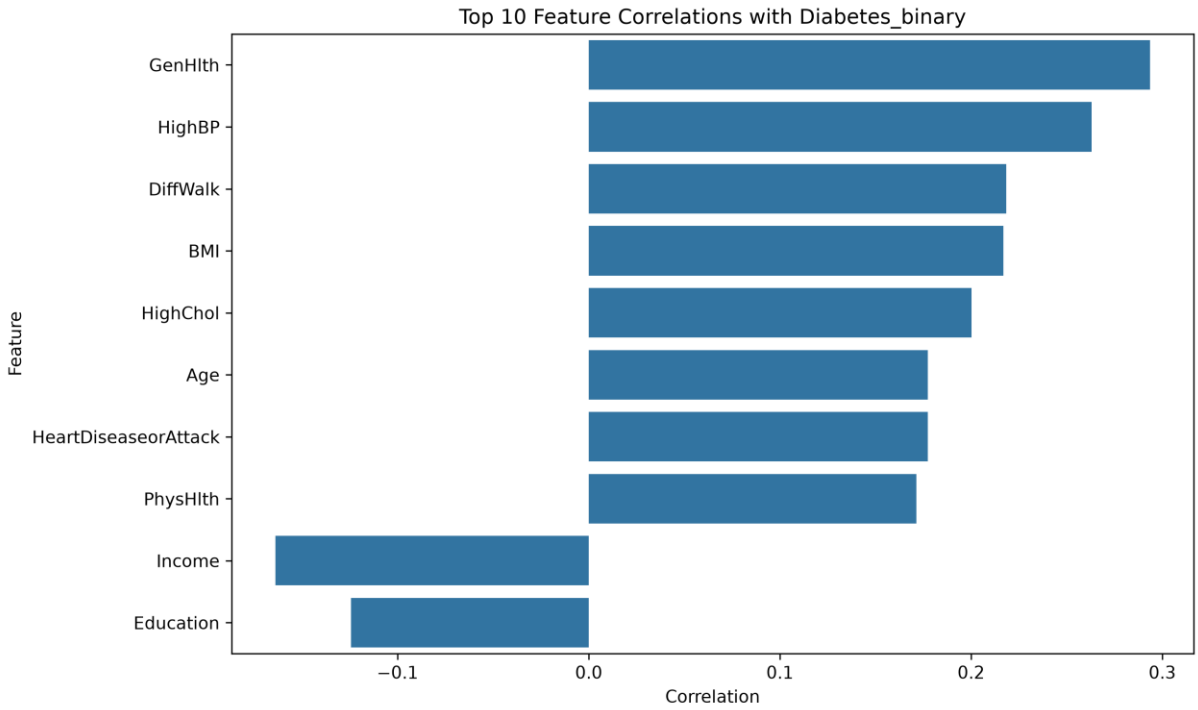


Figure 5. Features with the strongest linear association with the diabetes target.

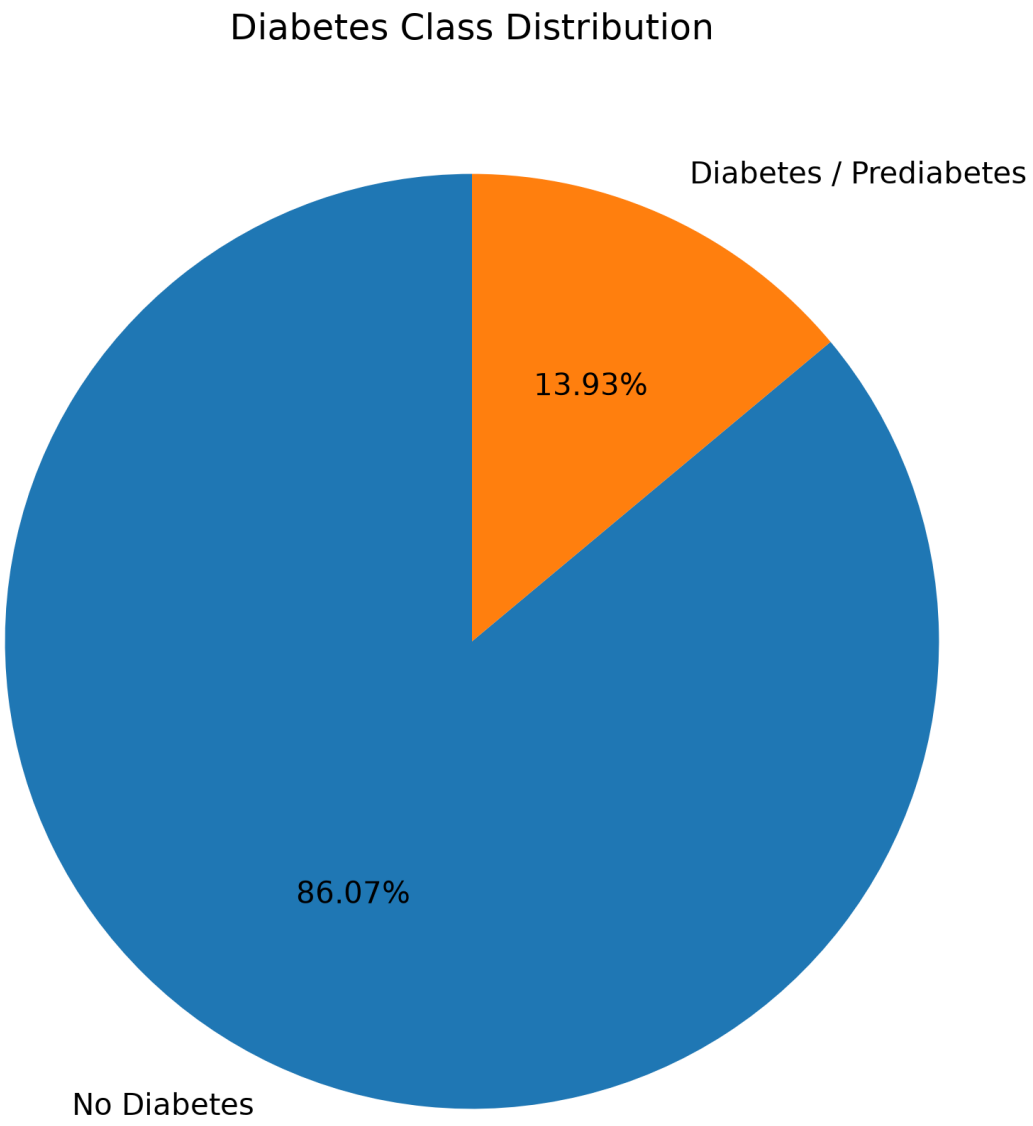


Figure 6. Proportion of diabetes and non-diabetes observations.

9. Class-wise Feature Analysis

Class-wise comparison was used to determine whether important features show different distributions between people classified as having diabetes and those classified as not having diabetes.

Feature	No Diabetes Mean	Diabetes Mean	Difference
BMI	27.806	31.944	+4.138
Income	6.19	5.21	-0.98

The diabetes group has a substantially higher average BMI in this dataset. Income also differs between groups, although this should be interpreted as an association in the survey data rather than a causal relationship.

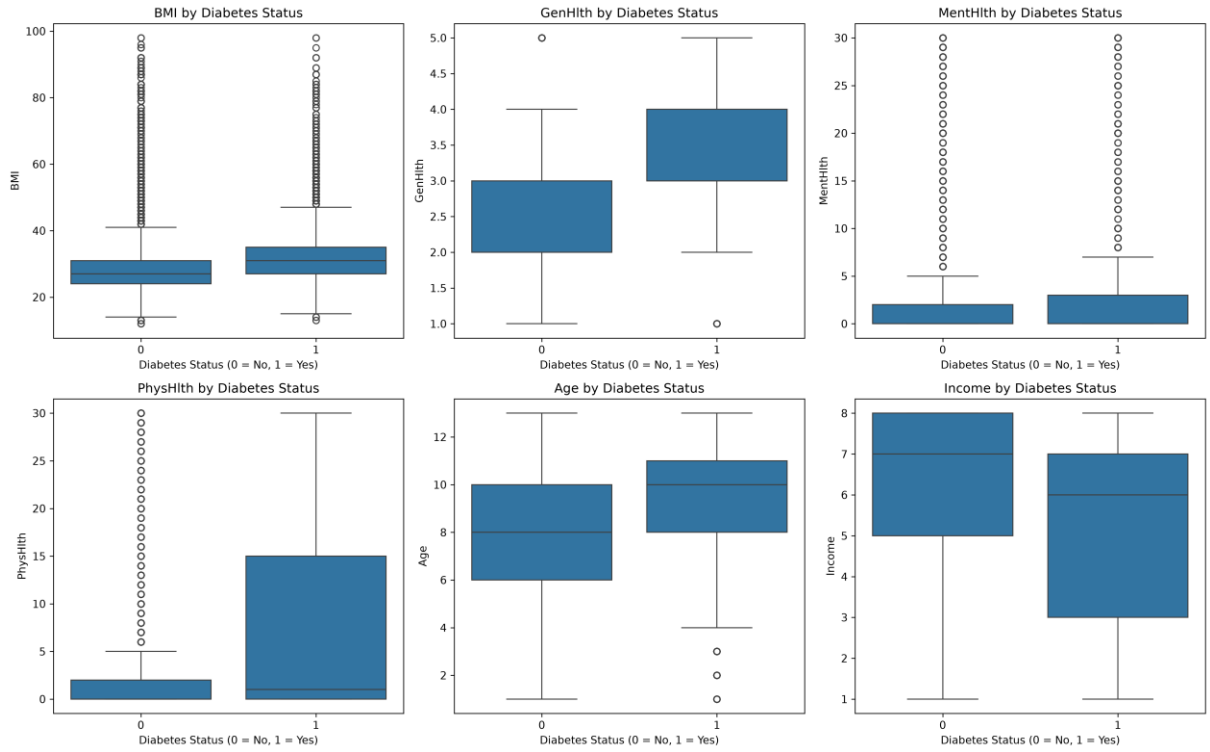


Figure 7. Class-wise comparison of selected features.

10. Diabetes Prevalence Analysis

Prevalence analysis provides a more interpretable view than raw correlation alone because it shows the percentage of diabetes-positive observations within categories.

10.1 General Health

GenHlth Category	Diabetes Prevalence
1	2.52%
2	7.16%
3	17.79%
4	31.01%
5	37.89%

Diabetes prevalence rises substantially across the GenHlth categories, from 2.52% in category 1 to 37.89% in category 5. This is one of the strongest descriptive patterns identified during EDA.

10.2 Age

Diabetes prevalence generally increases across the age categories, reaching approximately 21.85% at category 11 before slightly declining in the final categories.

10.3 Education and Income

The prevalence analysis shows an inverse descriptive relationship between diabetes prevalence and both Education and Income categories. For example, income-category prevalence decreases from 24.29% at category 1 to 7.96% at category 8.

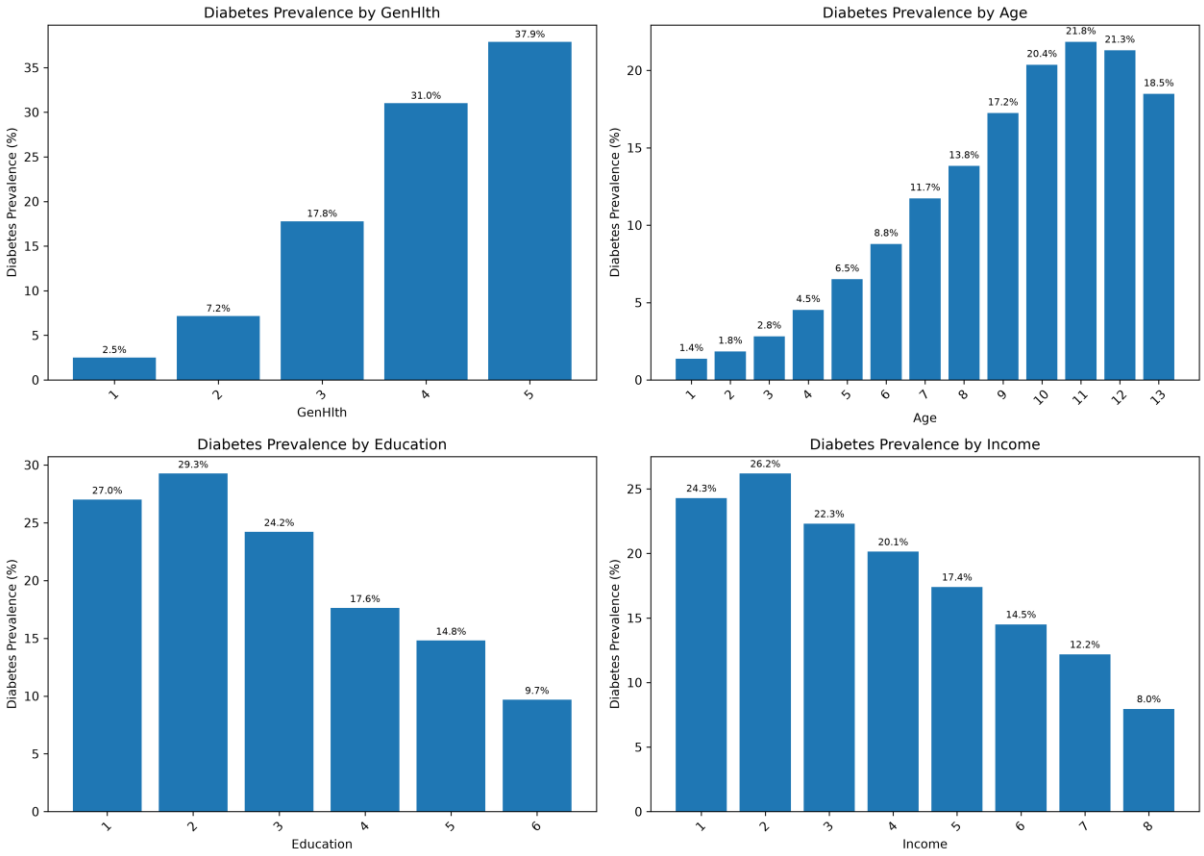


Figure 8. Diabetes prevalence across selected demographic, health and lifestyle features.

11. Binary Feature Analysis

Feature	Category 0	Category 1
HighBP	6.04%	24.45%
HighChol	7.98%	22.01%
HeartDiseaseorAttack	11.95%	32.97%
Stroke	13.18%	31.75%
DiffWalk	10.53%	30.75%
PhysActivity	21.14%	11.61%

Several health-related binary indicators show large differences in diabetes prevalence between their two categories. High blood pressure, high cholesterol, history of heart disease or attack, stroke and difficulty walking are associated with higher diabetes prevalence in this dataset. Physical activity shows the opposite descriptive pattern.

These findings are observational associations. They should not be interpreted as evidence that any individual factor directly causes or prevents diabetes.

12. Correlation Analysis

Pearson correlation was calculated between encoded numerical variables and the binary target. The strongest target correlations were:

Rank	Feature	Correlation with Target
1	GenHlth	0.293569
2	HighBP	0.263129
3	DiffWalk	0.218344
4	BMI	0.216843
5	HighChol	0.200276
6	Age	0.177442
7	HeartDiseaseorAttack	0.177282

8	PhysHlth	0.171337
9	Income	-0.163919
10	Education	-0.124456

The correlations are moderate rather than extremely strong. This supports the use of a nonlinear model because diabetes risk may depend on combinations and interactions among several features rather than a single variable.

13. Data Preprocessing

The preprocessing pipeline was designed to minimize leakage and maintain a clean separation between training and evaluation data.

1. Exact duplicate rows were removed.
2. The target variable Diabetes_binary was separated from the predictor matrix.
3. Features and target were split before model training.
4. Stratified train, validation and test partitions were created.
5. StandardScaler was fitted only on the training data.
6. The same fitted scaler was then applied to validation and test data.
7. The scaled arrays were converted into PyTorch tensors.

Fitting the scaler only on the training set is important because using validation or test information during preprocessing would introduce data leakage.

14. Train / Validation / Test Split

Partition	Rows	Purpose
Training	160,631	Model parameter learning
Validation	34,421	Model monitoring and early stopping
Test	34,422	Final unbiased evaluation

Stratification was used so that the relative proportion of the two target classes remained consistent across the partitions.

15. Handling Class Imbalance

The target distribution is highly imbalanced, with approximately 86.07% non-diabetes and 13.93% diabetes observations in the original dataset.

A model trained without imbalance awareness could obtain a seemingly high accuracy by favoring the majority class. To reduce this problem, a positive-class weight of approximately 5.5382 was used with BCEWithLogitsLoss.

This increases the training penalty for errors involving the minority diabetes class and encourages the model to identify more positive cases.

16. Deep Learning Model Architecture

The selected model is a fully connected Multi-Layer Perceptron (MLP). It was chosen because the project specifically focuses on deep learning and because an MLP can learn nonlinear relationships among the encoded health indicators.

Layer	Configuration
Input	21 features
Linear	21 → 64
Batch Normalization	64 units
Activation	ReLU
Dropout	30%
Linear	64 → 32
Batch Normalization	32 units
Activation	ReLU
Dropout	20%

Linear	32 → 16
Activation	ReLU
Output	16 → 1
Output activation	Sigmoid for probability

Batch normalization was used to stabilize intermediate representations, while dropout was used to reduce overfitting. ReLU provides nonlinear activation between the fully connected layers.

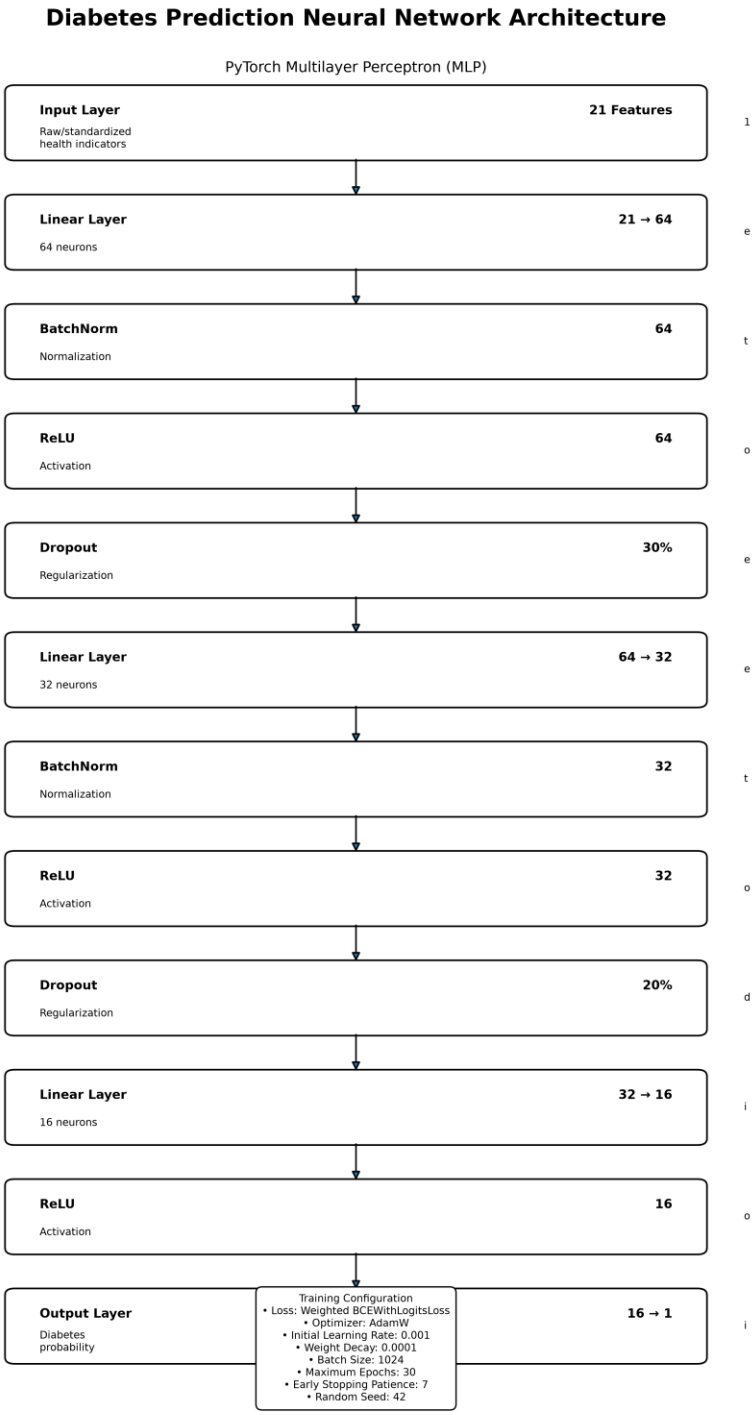


Figure 9. Neural-network architecture used for diabetes classification.

17. Model Training Configuration

Parameter	Value
Framework	PyTorch
Optimizer	AdamW
Learning rate	0.001
Weight decay	0.0001
Batch size	1024
Maximum epochs	30
Early stopping patience	7
Learning-rate reduction factor	0.5
Minimum learning rate	0.000001
Random seed	42
Loss function	Weighted BCEWithLogitsLoss
Training device	CPU

Early stopping was used to stop training when validation performance stopped improving. ReduceLROnPlateau was used to lower the learning rate when validation loss reached a plateau.

18. Training and Validation Performance

The best validation checkpoint occurred at epoch 13, with a validation loss of approximately 0.8748.

Metric	Validation
Accuracy	70.72%
Precision	31.76%
Recall	79.60%
F1-score	45.40%
ROC-AUC	0.8195
Best epoch	13
Validation loss	0.8748

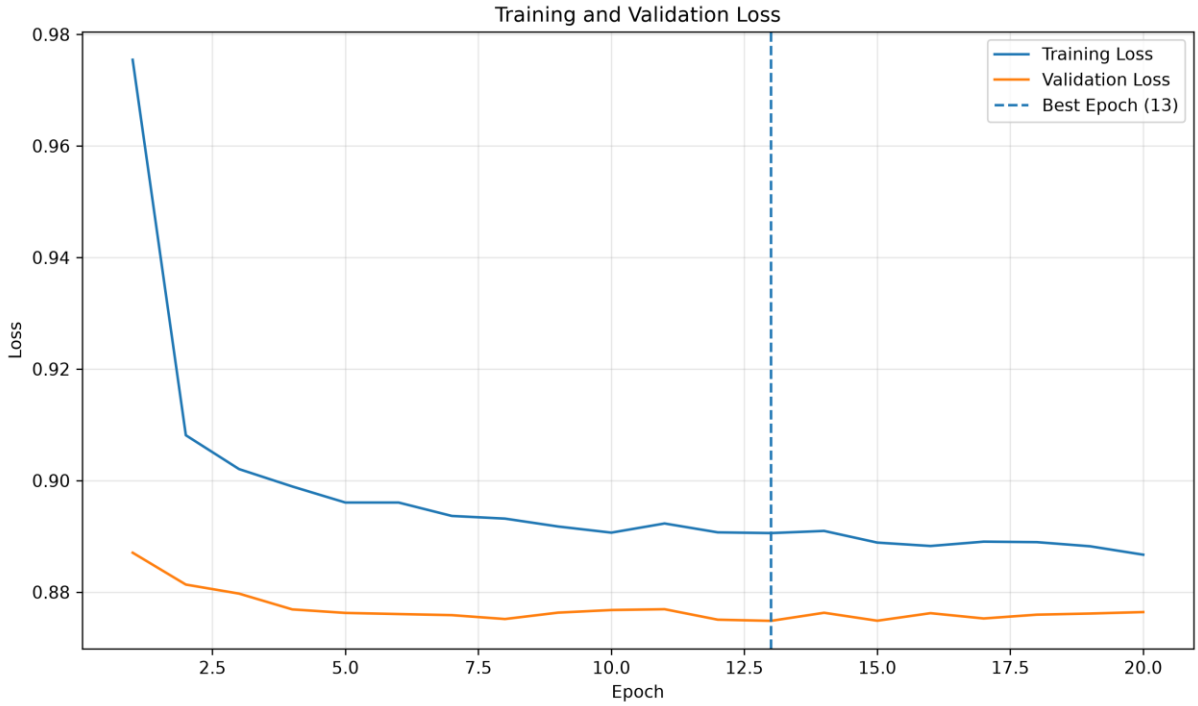


Figure 10. Training and validation loss during model training.

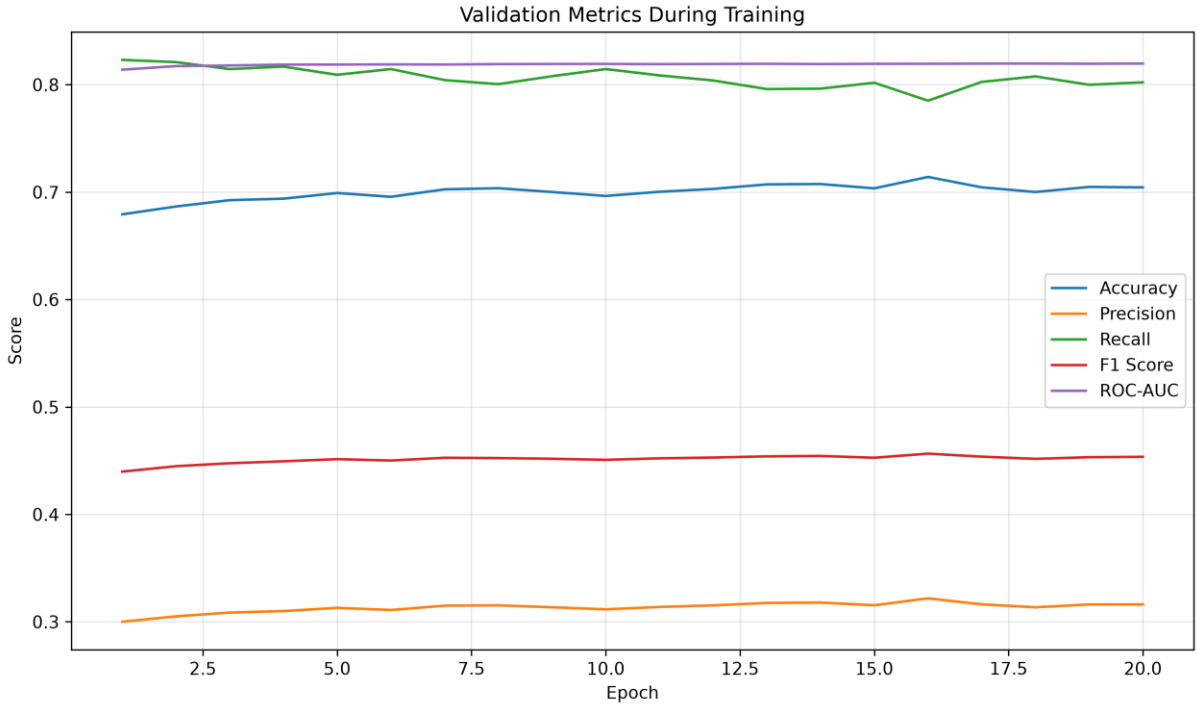


Figure 11. Validation classification metrics.

19. Test Set Evaluation

The final model was evaluated on the untouched test set after model development. At the default 0.50 probability threshold, the model achieved the following results:

Metric	Test Result
Accuracy	70.48%
Precision	31.54%
Recall	79.49%
F1-score	45.17%
ROC-AUC	0.8167

The ROC-AUC of 0.8167 indicates that the model has useful ranking ability across probability thresholds. The recall of 79.49% indicates that the model identifies a substantial proportion of diabetes-positive test cases, while the lower precision indicates a relatively high number of false-positive predictions.

20. Confusion Matrix and Error Analysis

	Predicted 0	Predicted 1
Actual 0	TN = 20,075	FP = 9,082
Actual 1	FN = 1,080	TP = 4,185

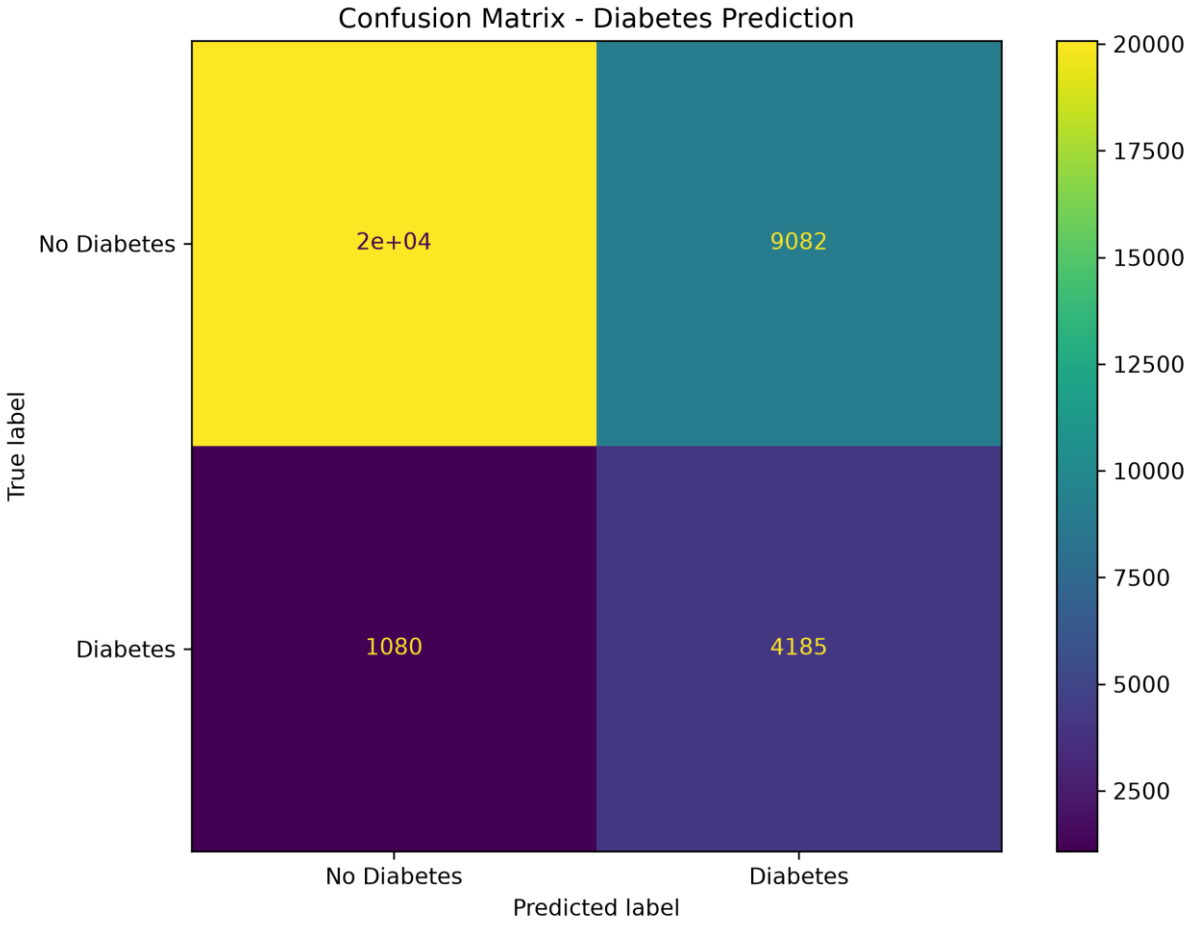


Figure 12. Test-set confusion matrix at the 0.50 threshold.

The model correctly identifies 4,185 diabetes-positive cases and misses 1,080 positive cases. However, it also produces 9,082 false-positive predictions.

The confusion matrix demonstrates why accuracy alone is insufficient. Although the overall accuracy is approximately 70.48%, the model's behavior is strongly influenced by the class imbalance and the selected decision threshold.

21. ROC-AUC Analysis

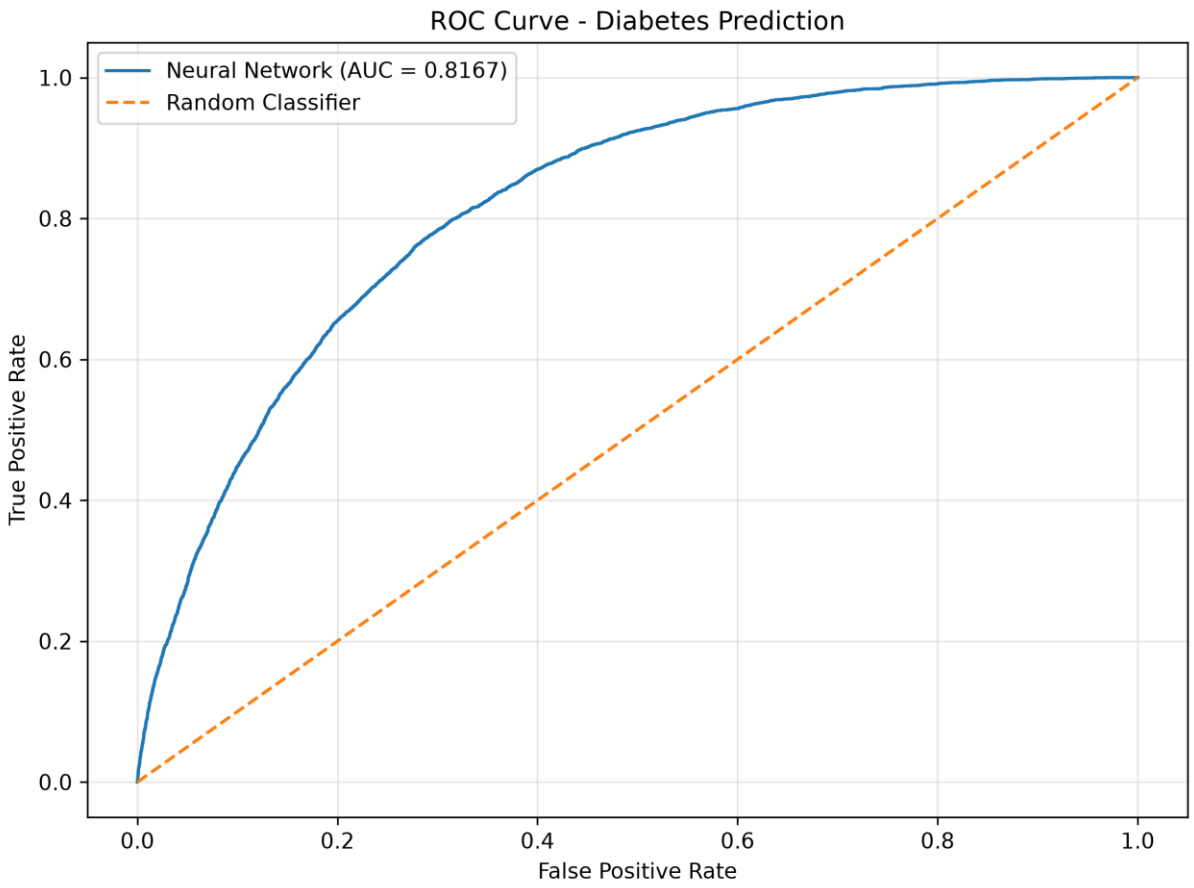


Figure 13. Receiver Operating Characteristic curve for the test set.

The test ROC-AUC is 0.8167. ROC-AUC evaluates how well the model ranks positive observations above negative observations across different classification thresholds. It is therefore useful when a single threshold may not be appropriate for every application.

22. Prediction Probability Distribution

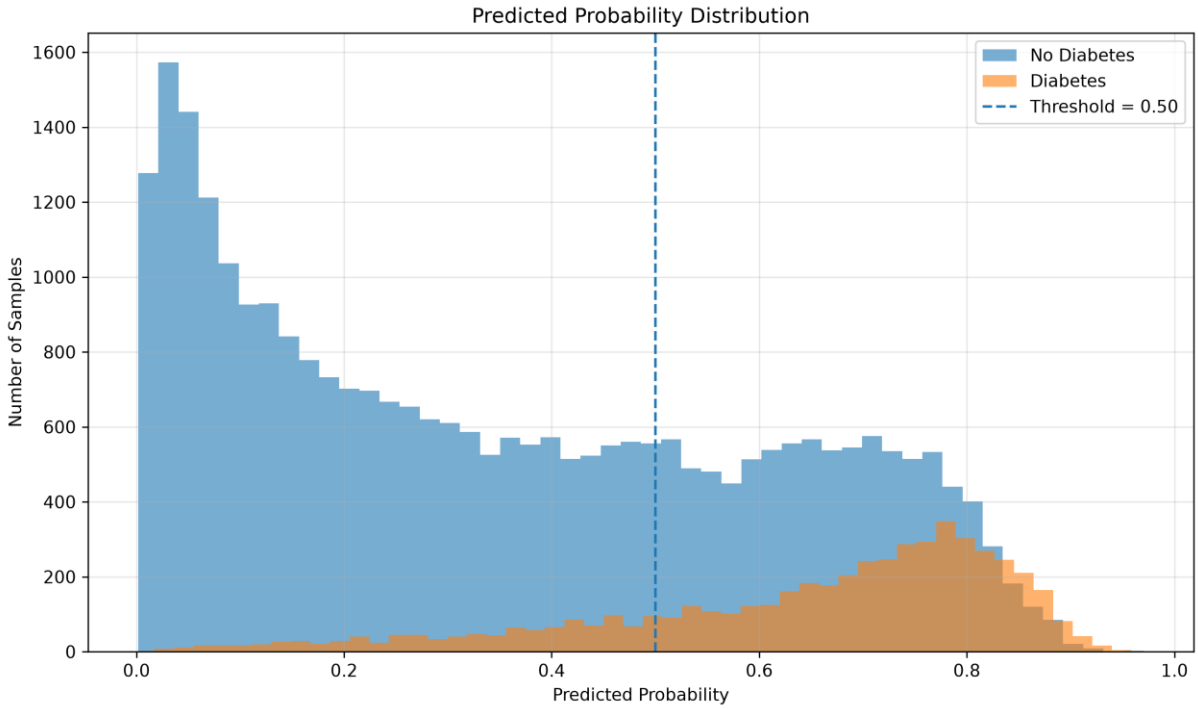


Figure 14. Distribution of predicted diabetes probabilities.

The probability distribution provides additional insight into how confidently the model separates the two classes. Overlap between the probability distributions is expected because health outcomes cannot be perfectly separated using the available survey indicators.

23. Probability Threshold Analysis

The default classification threshold of 0.50 is not necessarily the only reasonable operating point. A sensitivity analysis was performed to understand how changing the threshold affects precision, recall, accuracy and F1-score.

Threshold	Accuracy	Precision	Recall	F1
0.50	70.48%	31.54%	79.49%	45.17%
0.65	79.22%	38.59%	60.66%	47.17%

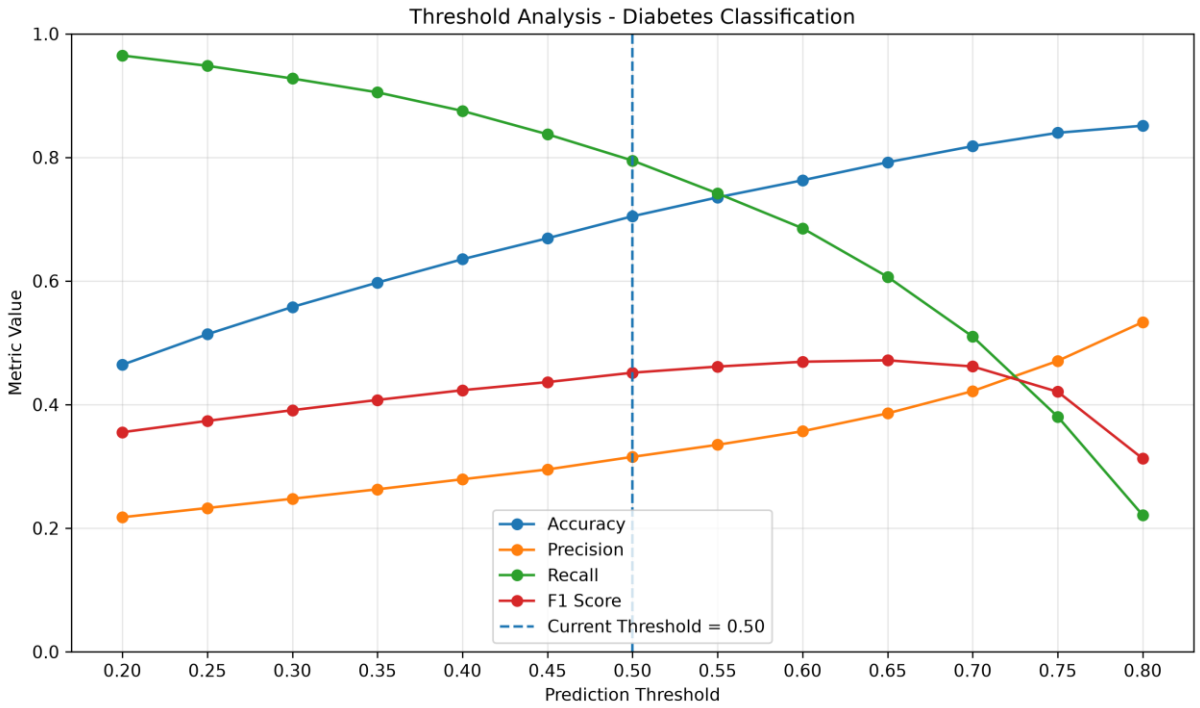


Figure 15. Sensitivity of classification metrics to the probability threshold.

At a threshold of 0.65, accuracy increases to 79.22% and precision increases to 38.59%, while recall decreases to 60.66%. The F1-score increases from 45.17% to 47.17%.

This analysis is presented as sensitivity analysis rather than as a production threshold-selection procedure. In a production workflow, the operating threshold should be selected using validation data and then evaluated once on an untouched test set.

24. Alternative Modeling Approaches

Model	Potential Advantage	Role in Future Work
Logistic Regression	Simple and interpretable baseline	Compare linear performance
Random Forest	Captures nonlinear feature interactions	Benchmark against tree-based learning
Gradient Boosting	Strong performance on tabular data	Compare advanced classical ML
MLP	Learns nonlinear neural representations	Primary deep learning model

The MLP was selected because the learning objective of the assignment is deep learning. However, a complete machine-learning study should also compare the neural model with strong traditional baselines.

25. Critical Analysis

25.1 Strengths

- The project performs detailed data-quality analysis before modeling.
- Exact duplicate observations were quantified and removed.
- Potential outliers were investigated rather than blindly deleted.
- Preprocessing avoids scaler leakage by fitting only on training data.
- Stratification preserves class proportions across data splits.
- Class imbalance is explicitly handled using weighted loss.
- Validation and test performance are reported using multiple metrics.
- Confusion-matrix analysis provides concrete error counts.

- Threshold sensitivity demonstrates the trade-off between precision and recall.
- The project includes visual and numerical evidence for major modeling decisions.

25.2 Areas for Improvement

- The model should be benchmarked against Logistic Regression and tree-based models.
- Hyperparameter tuning could be performed systematically using validation data.
- Cross-validation could provide a more robust estimate of model stability.
- Feature importance or explainability methods could improve interpretability.
- Probability calibration could be investigated before using model probabilities.
- The model could be evaluated across demographic subgroups for potential performance differences.

26. Limitations

- The dataset is survey-based and may contain self-reported information.
- The model learns associations in the dataset and cannot establish causality.
- The dataset is not a substitute for clinical diagnostic data.
- Class imbalance results in a precision-recall trade-off.
- The MLP was trained on CPU, which limits experimentation speed.
- The current project does not perform external clinical validation.
- The categorical and ordinal encodings may not perfectly represent the underlying clinical variables.
- Threshold selection for real-world deployment would require domain-specific validation and cost considerations.

27. Key Findings

- The dataset originally contained 253,680 records and 22 columns.
- There were 24,206 exact duplicate rows, representing 9.54% of the original dataset.
- After duplicate removal, 229,474 observations remained.
- The original diabetes-positive class represented 13.93% of observations.
- BMI, GenHlth, HighBP, HighChol and DiffWalk showed notable relationships with the target.
- Diabetes prevalence increased substantially across GenHlth categories.
- The MLP achieved a test ROC-AUC of 0.8167.
- At threshold 0.50, recall was 79.49%, while precision was 31.54%.
- The confusion matrix showed 4,185 true positives and 1,080 false negatives.
- Threshold analysis demonstrated a clear precision-recall trade-off.

28. Future Improvements

- Benchmark the MLP against Logistic Regression, Random Forest and Gradient Boosting.
- Perform systematic hyperparameter optimization.
- Experiment with class-weighting versus alternative imbalance strategies.
- Investigate calibrated probabilities.
- Add SHAP or permutation-based explainability.
- Evaluate subgroup-level performance.
- Perform repeated cross-validation.
- Investigate feature selection and dimensionality reduction.
- Experiment with more advanced tabular deep-learning architectures.
- Package the model into a simple educational prediction interface.

29. Technologies Used

Technology	Purpose
Python	Programming language
Pandas	Data loading and analysis
NumPy	Numerical operations
Matplotlib	Visualization
Seaborn	Statistical visualization
Scikit-learn	Preprocessing and evaluation
PyTorch	Deep learning model
UCI Repository	Dataset source
VS Code	Development environment

30. Project Structure

```
Week_5_Deep_Learning/
│
├── data/
│   └── diabetes_binary_health_indicators_BRFSS2015.csv
│
├── models/
│   ├── deep_eda_binary_feature_prevalence.csv
│   ├── deep_eda_class_comparison.csv
│   ├── deep_eda_correlation_ranking.csv
│   ├── deep_eda_duplicate_impact.csv
│   ├── deep_eda_numeric_summary.csv
│   ├── deep_eda_outlier_analysis.csv
│   ├── deep_eda_prevalence_age.csv
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│   ├── 05_class_distribution_pie.png
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│   ├── 06_highbp_vs_diabetes.png
│   ├── 06_highchol_vs_diabetes.png
│   ├── 06_physHlth_vs_diabetes.png
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├── src/
│   ├── 08_deep_eda_analysis.py
│   └── 09_create_final_report.py
│
```

```
├── requirements.txt
├── README.md
└── .gitignore
```

31. Learning Outcomes

- Understanding binary classification with neural networks.
- Understanding class imbalance and weighted loss functions.
- Applying data preprocessing without leakage.
- Using train, validation and test datasets correctly.
- Building an MLP using PyTorch.
- Understanding forward propagation and nonlinear activations.
- Using BatchNorm and Dropout for model regularization.
- Using AdamW optimization.
- Applying early stopping and learning-rate scheduling.
- Evaluating classification models beyond accuracy.
- Interpreting confusion matrices and ROC-AUC.
- Understanding threshold-based precision-recall trade-offs.
- Performing deeper exploratory data analysis before modeling.

32. Conclusion

This project demonstrates a complete deep learning workflow for binary classification using a health-indicator dataset. The workflow extends from data-quality assessment and exploratory analysis to preprocessing, neural-network design, training, evaluation and critical interpretation.

The analysis identified substantial class imbalance and a meaningful number of exact duplicate records. It also demonstrated why IQR-based outlier detection must be interpreted according to the semantics of the underlying variables. Rather than deleting valid health-survey observations solely because they were statistically unusual, the project documented their potential impact and retained them.

The final MLP achieved a test ROC-AUC of 0.8167, with 79.49% recall and 31.54% precision at the 0.50 threshold. These results demonstrate useful predictive separation while also highlighting the practical trade-off between identifying positive cases and generating false positives.

Overall, the project provides an evidence-based deep learning workflow and a strong foundation for future comparison with traditional machine learning models, explainability methods and more advanced tabular deep-learning techniques.

33. Disclaimer

This project is intended for educational and academic purposes only. The predictions and statistical associations produced by the model should not be interpreted as medical diagnosis, clinical advice or a substitute for evaluation by a qualified healthcare professional.